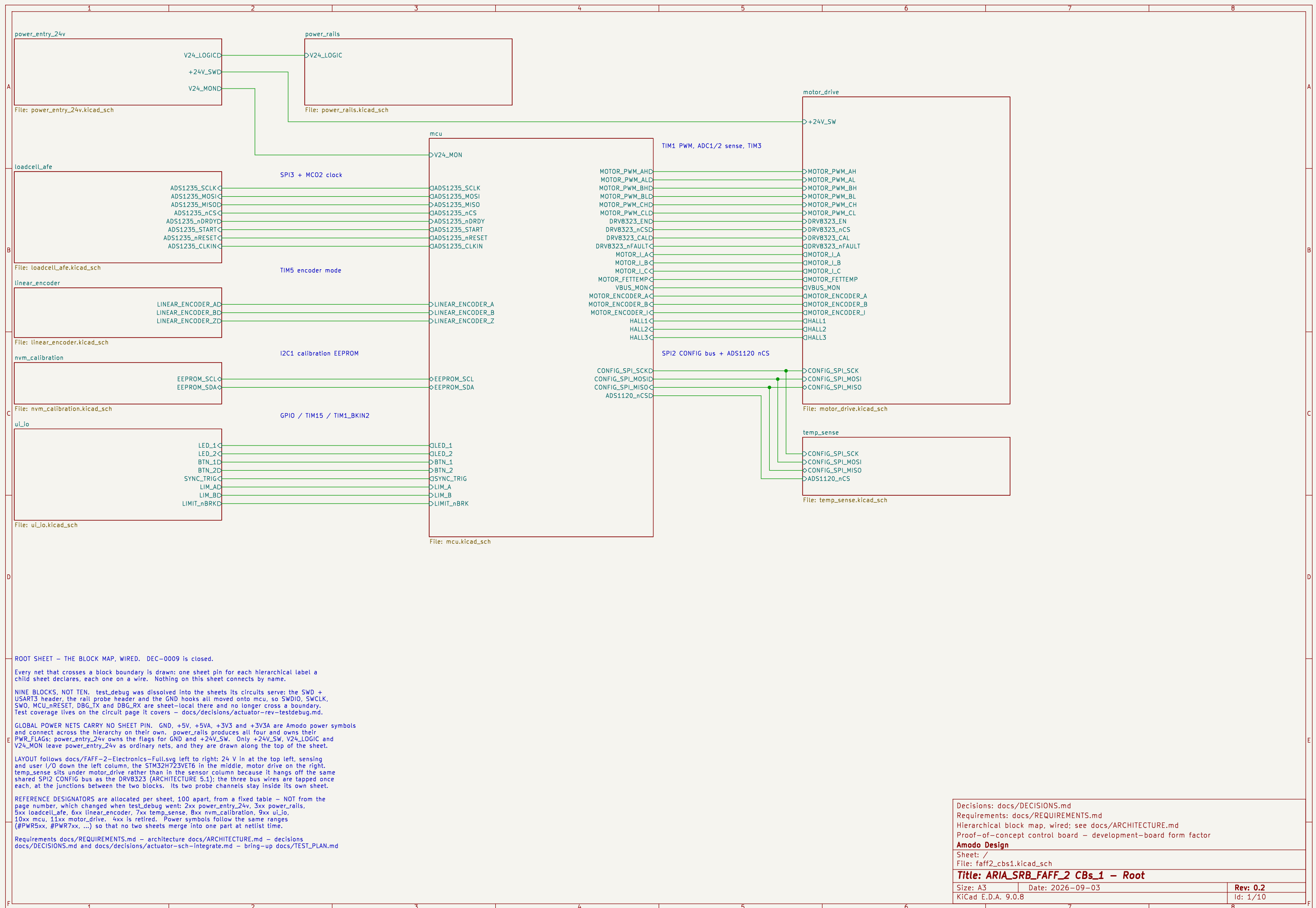
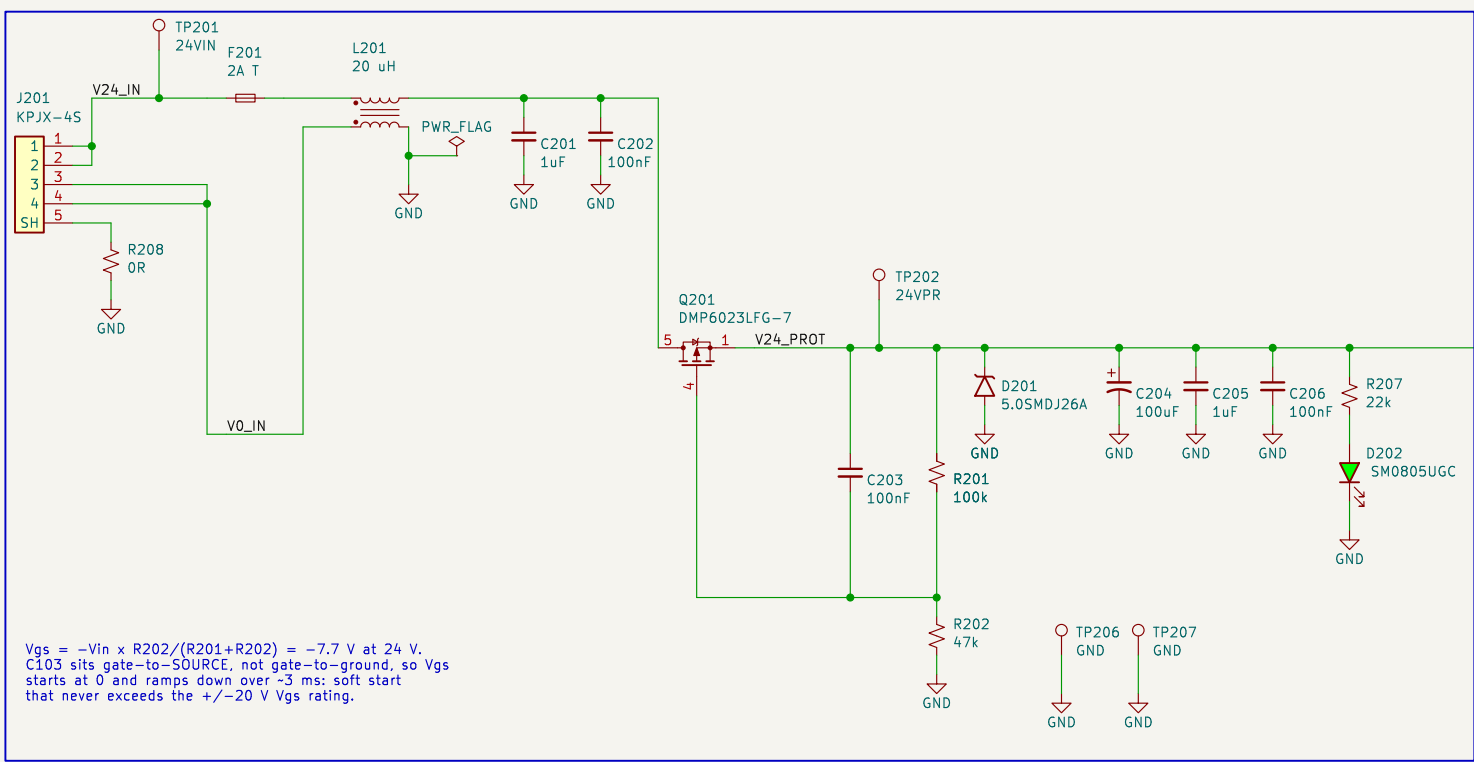
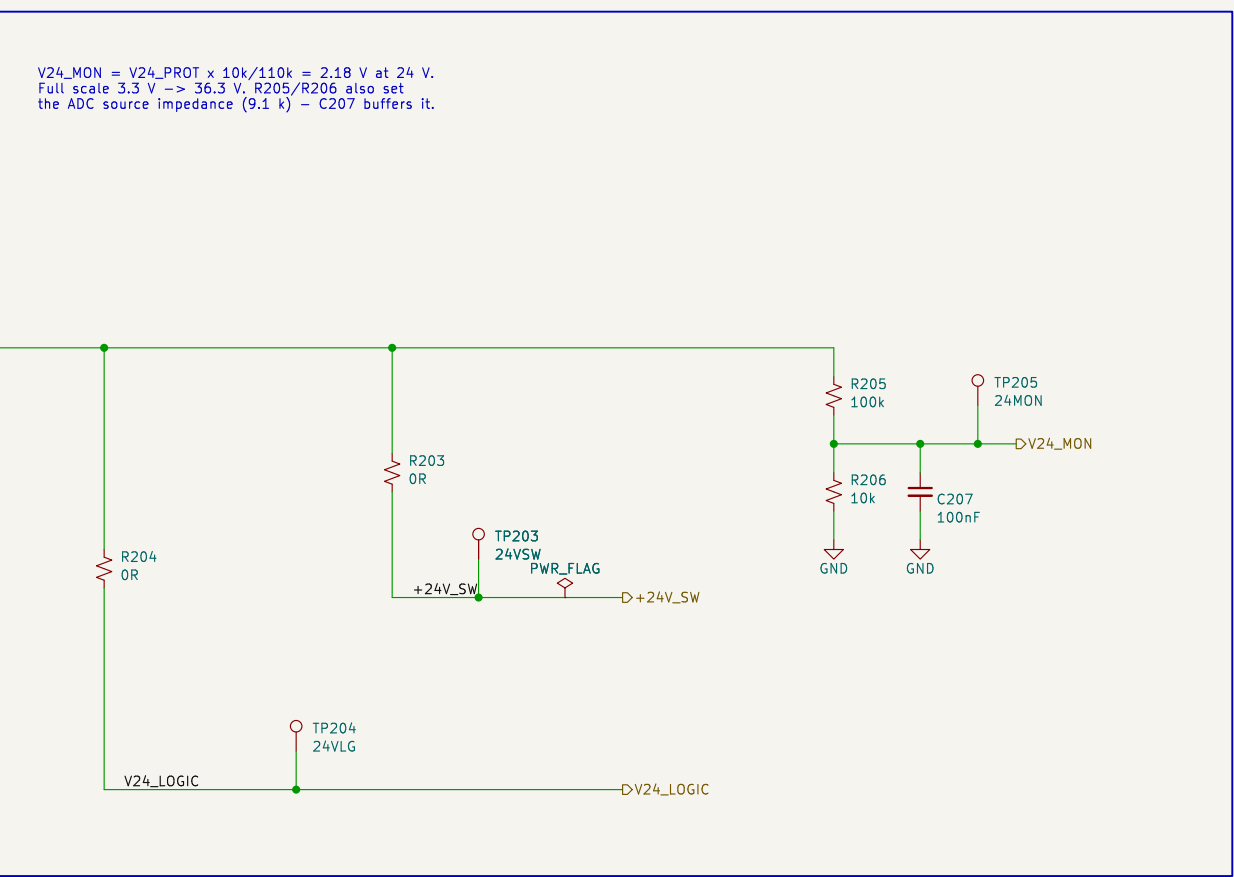




A. 24 V INPUT, FUSING, CM FILTER AND REVERSE-POLARITY PROTECTION



B. BRANCH ISOLATION LINKS / CURRENT BREAKS, RAIL MONITOR AND TEST POINTS



BLOCK: power\_entry\_24v - 24 V DC input and protection (REQ-EL-01..04)

CHAIN J201 KPJX-4S (contacts doubled: 1,2 = +24 V, 3,4 = 0 V, 5 = shell)  
-> F201 2 A time-lag (25 W peak = 1.04 A at 24 V; 2 A T holds 1.5 A)  
-> L201 common-mode choke, both lines; board GND is defined on its load side  
-> Q201 P-channel reverse-polarity series switch (drain = supply, source = load, so the body diode blocks a reversed supply and nothing downstream conducts)  
-> D201 TVS + C204..C206 bulk/HF -> V24\_PROT, the protected 24 V bus.

BRANCHES (docs/TEST\_PLAN.md 3.3 - each an isolation link AND a current break)  
R203 -> +24V\_SW : motor\_drive only. Open R203 and the whole board can be exercised with no possibility of the actuator moving.  
R204 -> V24\_LOGIC: power\_rails only. Independent of R203.  
One OR link per branch serves as both the isolation link and the current break; a second series part would add joints and no capability.

GROUND One GND net board-wide. AGND / DGND / PGND in the block stub notes are layout partitions of it, not separate nets - see the decisions file.  
This sheet owns the single GND PWR\_FLAG; no other sheet may add one.

INTERFACES exported (hierarchical labels, wired to root sheet pins - DEC-0009)  
OUT +24V\_SW -> motor\_drive OUT V24\_LOGIC -> power\_rails  
OUT V24\_MON -> mcu PC1 / ADC1\_INP11  
Global power nets used here: GND.

Reasoning, part sizing and residual ERC: docs/decisions/actuator-sch-power.md

Design reasoning: docs/decisions/actuator-sch-power.md  
Requirements: REQ-EL-01..04 - docs/REQUIREMENTS.md  
Independent motor / logic branch links (TEST\_PLAN 3.3)  
24 V in on KPJX-4S: fuse, CM choke, reverse polarity, TVS

Amodo Design

Sheet: /power\_entry\_24v/  
File: power\_entry\_24v.kicad\_sch

Title: CBs\_1 - Power Entry 24 V

|                    |                  |          |
|--------------------|------------------|----------|
| Size: A3           | Date: 2026-09-02 | Rev: 0.2 |
| KiCad E.D.A. 9.0.8 |                  | Id: 2/10 |

**A. PRE-REGULATOR V24\_LOGIC -> 5.5 V (LMR33630, 400 kHz)**

**B. POST-REGULATED 5 V RAILS (ADPL42005: +5V digital, +5VA analog)**

**B. POST-REGULATED 5 V RAILS (ADPL42005: +5V digital, +5VA analog)**

EN is tied to each LDO input – the datasheet's own automatic-startup connection – so the 5 V rails follow the pre-regulator with no sequencing logic. SENSE ties to VOUT at the regulator, not past the OR link, so lifting the link cannot open the loop. PG is unused; no-connect, not tied to RAIL\_PG00D. +5VA feeds ADS1235 / ADS1120 AVDD and the AFE excitation chain; +5V feeds the IKP11 read head.

The schematic diagram illustrates the LMR33630 buck converter circuit. The U304 LMR33630 IC is the central component, with its pins connected as follows:

- VIN (Pin 2):** Connected to the V24\_LOGICD input.
- EN (Pin 3):** Connected to the V24\_LOGICD input.
- PGND (Pin 1):** Connected to GND.
- VCC (Pin 6):** Connected to GND.
- TP (Pin 7):** Connected to GND.
- BOOT (Pin 7):** Connected to the SW pin (Pin 8).
- SW (Pin 8):** Connected to the SW pin (Pin 8).
- PG (Pin 4):** Connected to the SW pin (Pin 8).
- FB (Pin 5):** Connected to the FB pin (Pin 5).
- FB (Pin 5):** Connected to the FB pin (Pin 5).

The circuit includes several passive components:

- Resistors:** R308 (100k), R309 (8.2k), R310 (100k), R311 (43k), R312 (0R), R316 (1k).
- Capacitors:** C315 (10uF), C316 (10uF), C317 (220nF), C318 (1uF), C319 (100nF), C320 (22uF), C321 (22uF), C322 (100nF).
- Inductor:** L302 (6.8uH).

The output of the converter is connected to the +3V3 rail, which is also connected to the TP306 +3V3 test point. The PWR\_FLAG pin (Pin 12) is connected to the +3V3 rail.

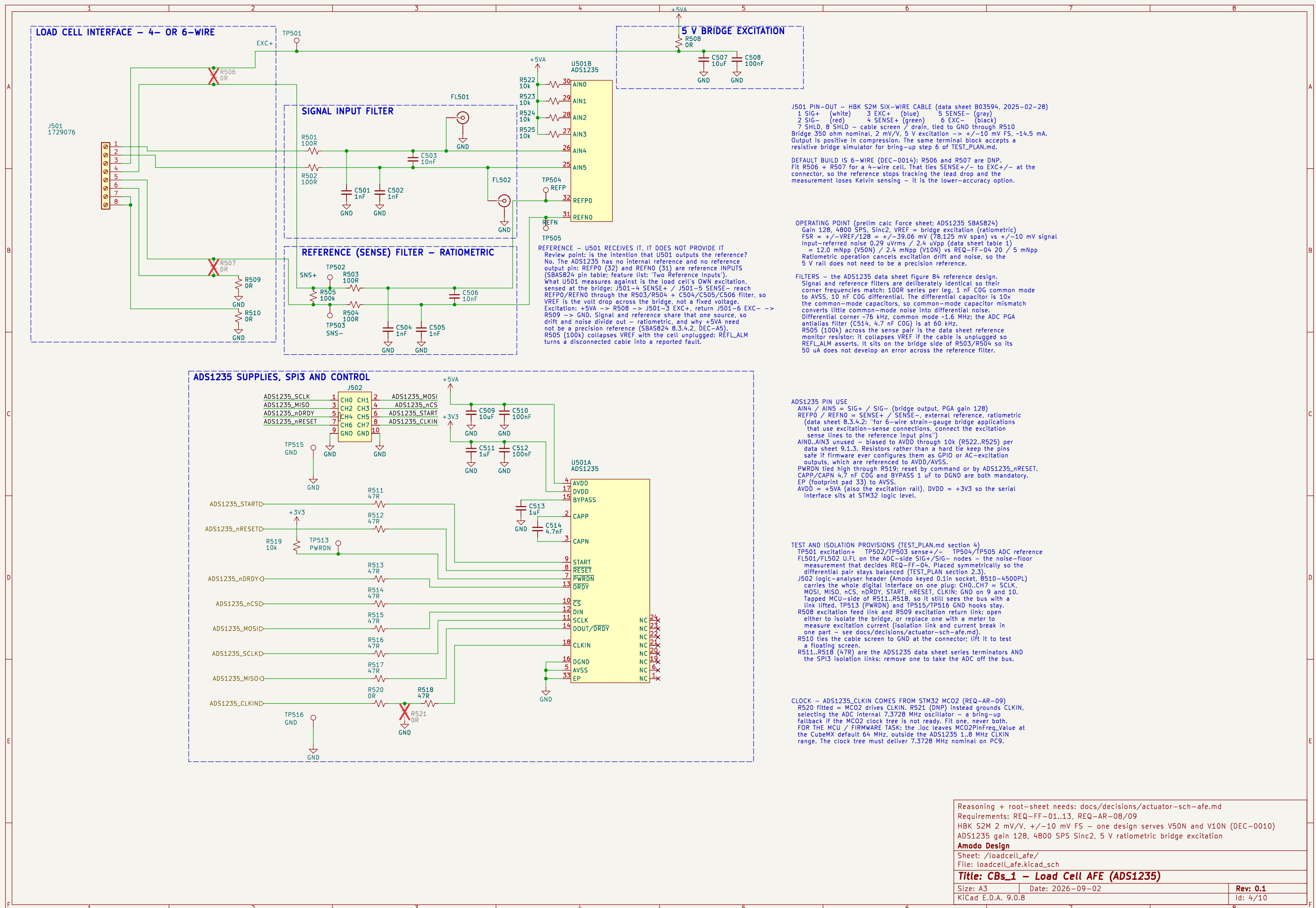
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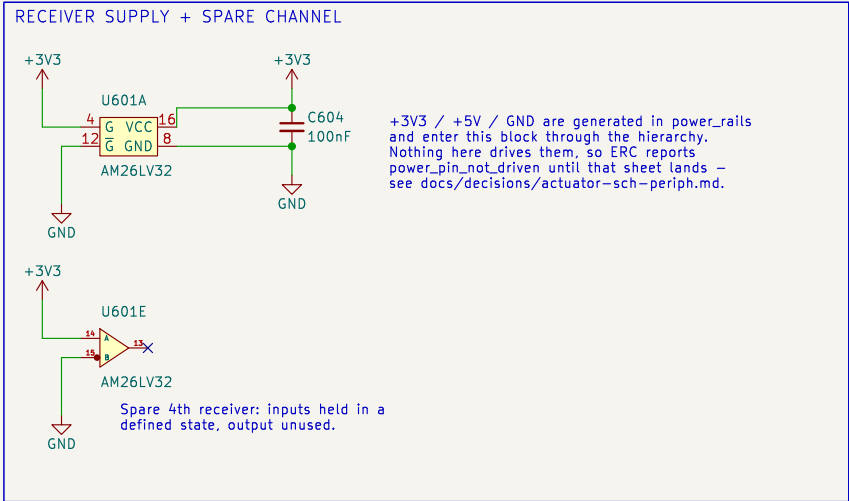
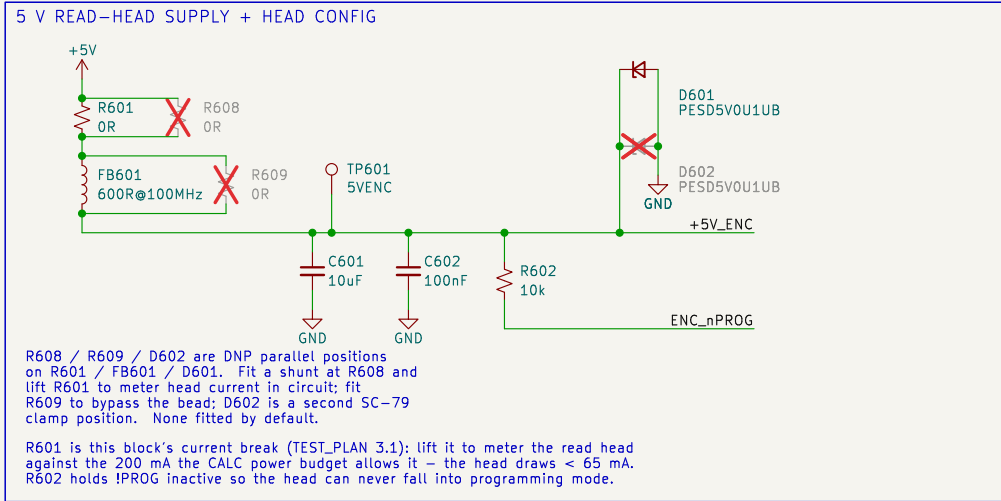
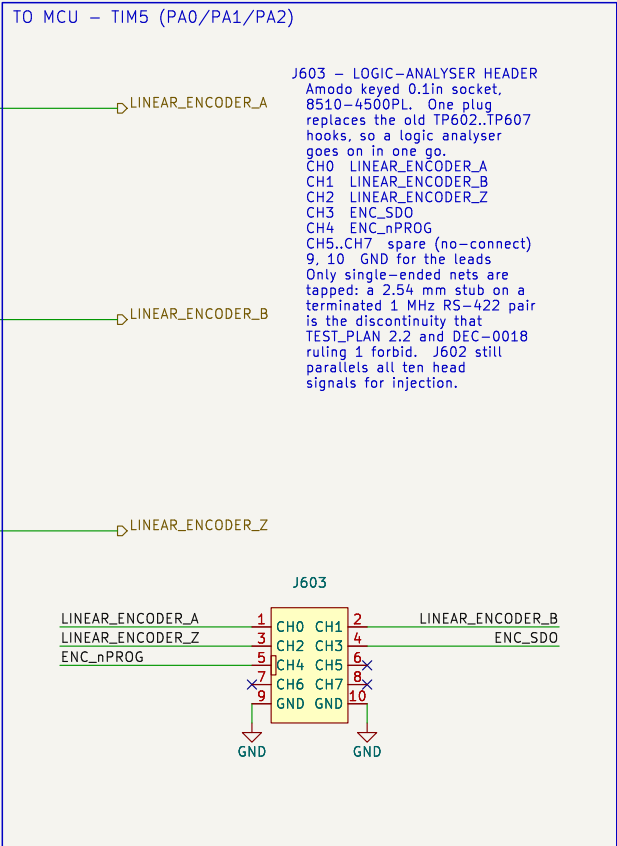
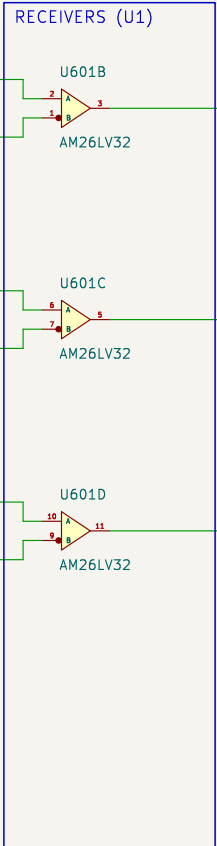
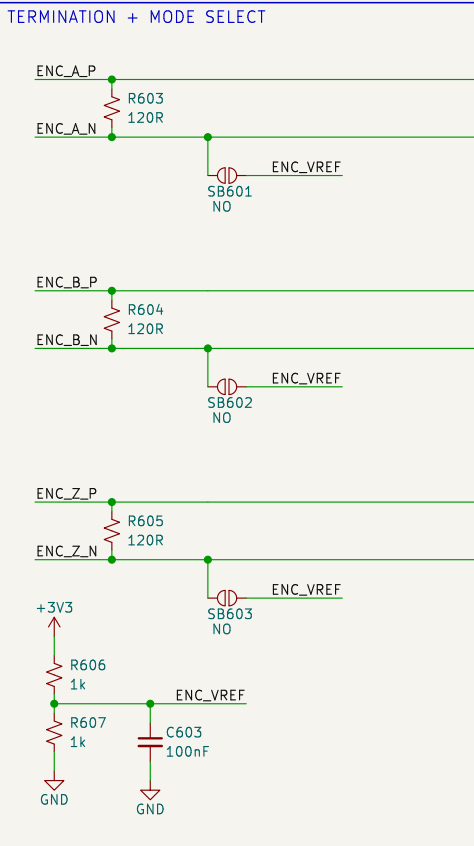
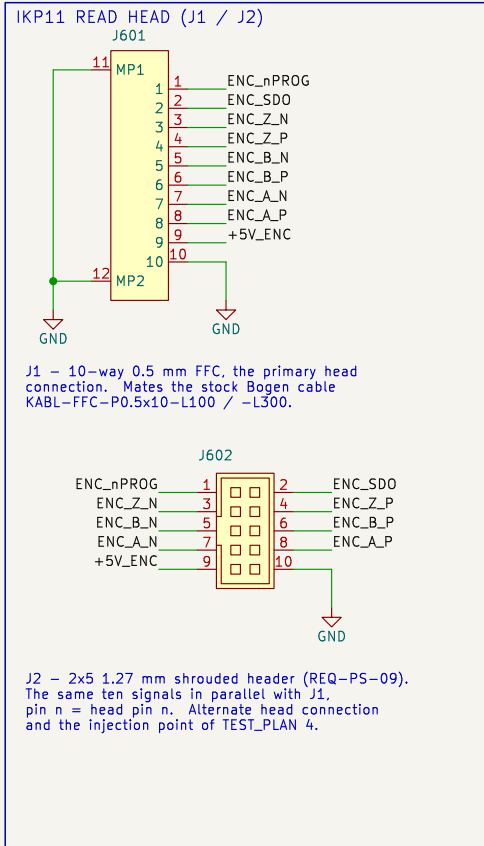
|       |       |                                                                                                                                                            |
|-------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RAILS | +5V5  | LMR33630 buck from V24_LOGIC. Pre-regulator only, local to this sheet.                                                                                     |
|       | +5V   | ADPL42005 LDO from +5V5. IKP11 read head (<65 mA plus RS-422 drive) and spare.                                                                             |
|       | +5VA  | ADPL42005 LDO from +5V5. ADS1235 / ADS1200 AVDD and the AFE 5 V excitation and reference chain, which lives in loadcell-Lafe - take it from +5VA, not +5V. |
|       | +3V3  | LMR33630 buck from V24_LOGIC. S3132, USB3320, DRV8323 logic, QSPI, EEPROM, RS-422 receiver, rotary encoder.                                                |
|       | +3V3A | -1.1 A worst case against a 3 A part.<br>ferrite-isolated from +3V3 for the ADC digital supplies.                                                          |

|   |   |
|---|---|
| 1 | 2 |
|---|---|

Reasoning, part sizing and residual ERC:  
docs/decisions/actuators-sch-power.md

|                    |                  |          |
|--------------------|------------------|----------|
| Size: A3           | Date: 2026-09-02 | Rev: 0.2 |
| KiCad E.D.A. 9.0.8 |                  | Id: 3/10 |





#### BLOCK NOTES - linear\_encoder

Ordered head: Bogen IKP11-Z1.4-P1-V5-D1-R0.5-F1000-C1 (DEC-0003, REQ-PS-05)  
Z1.4 periodic index from the pole pitch, 4 counts P1 1 mm pole pitch  
V5 5 V supply, < 65 mA D1 RS-422 A/B/Z  
R0.5 0.5 um resolution, post-quadrature F1000 1 MHz per channel  
C1 15.8 x 15.4 PCB: a 10-way 0.5 mm FFC connector AND a 1.27 mm pad row

HEAD PIN ORDER (datasheet rev 3.7 p.6 - same order on the FFC and the pads)  
1 !PROG 2 SDO 3 /Z 4 Z 5 /B 6 B 7 /A 8 A 9 V+ 10 V-  
Check the FFC cable contact side (type A / type D) against J1 before ordering - J1 is a bottom-contact receptacle. J2 needs no such check and is the fallback.

#### MODES

RS-422, the ordered D1 build (default): R603/R604/R605 fitted, SB601/SB602/SB603 open.  
120 R across each pair, as the head datasheet asks (ZO = 120 R at the receiving end). No external fail-safe bias is needed: the AM26LV32 has open-circuit, shorted AND 100 R-terminated fail-safe, so an unplugged or dead head drives a static high - a stalled count, never phantom motion.  
Single-ended TTL / bench injection: remove R603/R604/R605, close SB601/SB602/SB603.  
Each inverting input then sits on ENC\_VREF (1.65 V), so a 0-3.3 V or 0-5 V single-ended source on ENC\_XP decodes correctly. This covers a TTL head variant (order code D3) and injection through J2 with no read head fitted.

U1 runs from 3V3, so its outputs are native 3V3 logic (REQ-PS-06); its -0.3 V to +5.5 V input common-mode range accepts the 5 V head directly, and it switches to 32 MHz against the 1 MHz per channel of REQ-PS-07. Both enables are tied active: G (pin 4) high, /G (pin 12) low.

No test points sit on the differential pairs: a probe stub on a terminated 1 MHz RS-422 line is a discontinuity (TEST\_PLAN 2.2), and the receiver outputs carry the same information single-ended. J603, the keyed logic-analyser socket, replaces the old TP602.TP607 hooks: CH0..CH4 = LINEAR\_ENCODER\_A / \_B / \_Z, ENC\_SDO, ENC\_nPROG, with GND on 9 and 10 for the ground leads. TP601 stays, a 1.0 mm SMT pad for a DMM on the 5 V head supply.

Test strategy: docs/TEST\_PLAN.md  
REQ-PS-01..10, REQ-AR-07, DEC-0003  
RS-422 A/B/Z -> 3V3 into TIM5 (encoder mode TI12 + CH3 capture)  
Bogen IKP11-Z1.4-P1-V5-D1-R0.5-F1000-C1 read head interface

#### Amodo Design

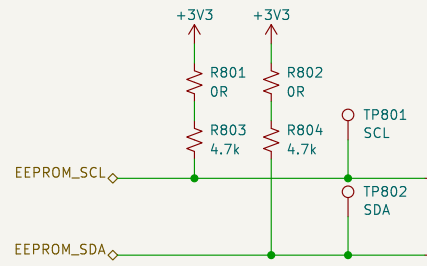
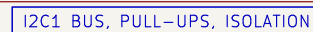
Sheet: /linear\_encoder/  
File: linear\_encoder.kicad\_sch

#### Title: CBs\_1 - Linear Encoder (IKP11)

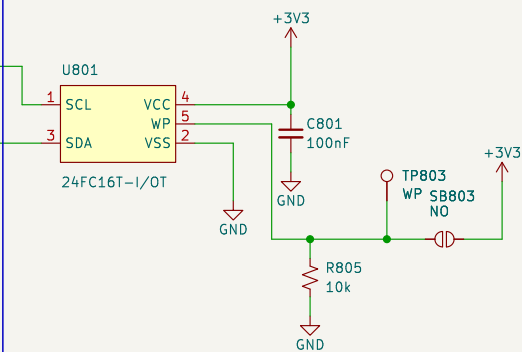
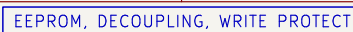
Size: A3 Date: 2026-09-02 Rev: 0.2  
KiCad E.D.A. 9.0.8 Id: 5/10





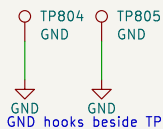


R801/R802 are the removable links of TEST\_PLAN 4: lift either to drop its pull-up. SB801/SB802 are cut to take the EEPROM off I2C1 while the bus stays pulled up – this block owns that break, so mcu must not duplicate it. The SDA pull-up crosses the SCL run, no junction.



WP is ACTIVE HIGH. R805 holds it low, so the part is writable and firmware can store calibration at any time; solder SB803 to lock the device permanently. TP3 lets a jig drive WP either way. This resolves OQ-06 – no MCU pin is spent on write protect.

## GROUND HOOKS



+3V3 and GND are generated in power\_rails and enter this block through the hierarchy; nothing here drives them, so ERC reports power\_pin\_not\_driven until that sheet lands. See docs/decisions/actuator-sch-periph.md.

analyse clips.

## BLOCK NOTES – nvm\_calibration

U1 = Microchip 24FC16T-I/OT, 16 kbit (2048 x 8) I2C EEPROM in SOT-23-5.

Pin map verified against the datasheet package drawing (DS20001703R p.1):

1 SCL 2 VSS 3 SDA 4 VCC 5 WP.  
1 7-5.5 V and 1 MHz capable, so it keeps

1.7–5.5 V and 1 MHz capable, so it keeps up with I2C1 at either 400 kHz or Fast-mode Plus, and 4 million erase/write cycles covers repeated calibration.

2 KB is sized for coefficients, not tables: force and temperature calibration constants, the variant (V50N / V10N – see DEC-0010, where the variants differ only by a calibration constant), board serial number and build state. Bulk force profiles live in the OCTOSP11 RAM on the mcu sheet (DEC-0006), not here.

**ADDRESSING CAVEAT.** The 24xx16 uses all three block-select bits internally, so it answers to the **WHOLE** 1010xxx address space and A0/A1/A2 do not exist on the part. No second 1010-family device can share I2C1. It is the only device on the bus today; if another is ever added, swap to a smaller 24xx02/04 first.

4k7 pull-ups suit 400 kHz on a short board-level bus with one load. They are the only pull-ups on I2C1: the mcu block must not add its own.

Test strategy: docs/TEST\_PLAN.md  
REQ-EL-08, REQ-AR-11; resolves OQ-06 (write protect)  
I2C1: EEPROM\_SCL PB8, EEPROM\_SDA PB9  
16 kbit I2C EEPROM for calibration and compensation data

**Amodo Design**

Sheet: /nvm\_calibration/  
File: nvm\_calibration.kicad\_sch

**Title: CBs\_1 – NVM / Calibration Store**

Size: A3

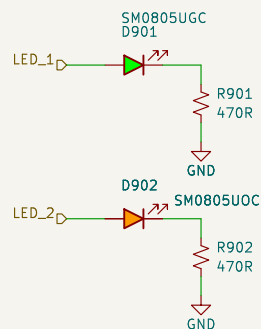
Date: 2026-09-02

Rev: 0.2

KiCad E.D.A. 9.0.8

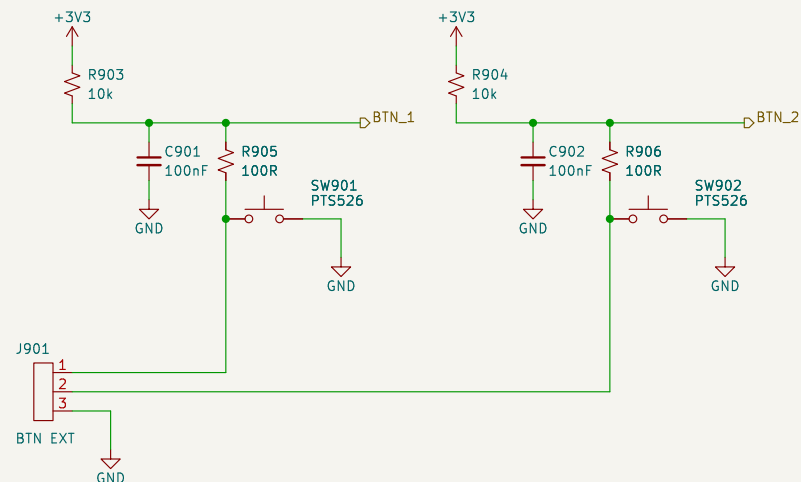
Id: 7/10

## STATUS LEDS



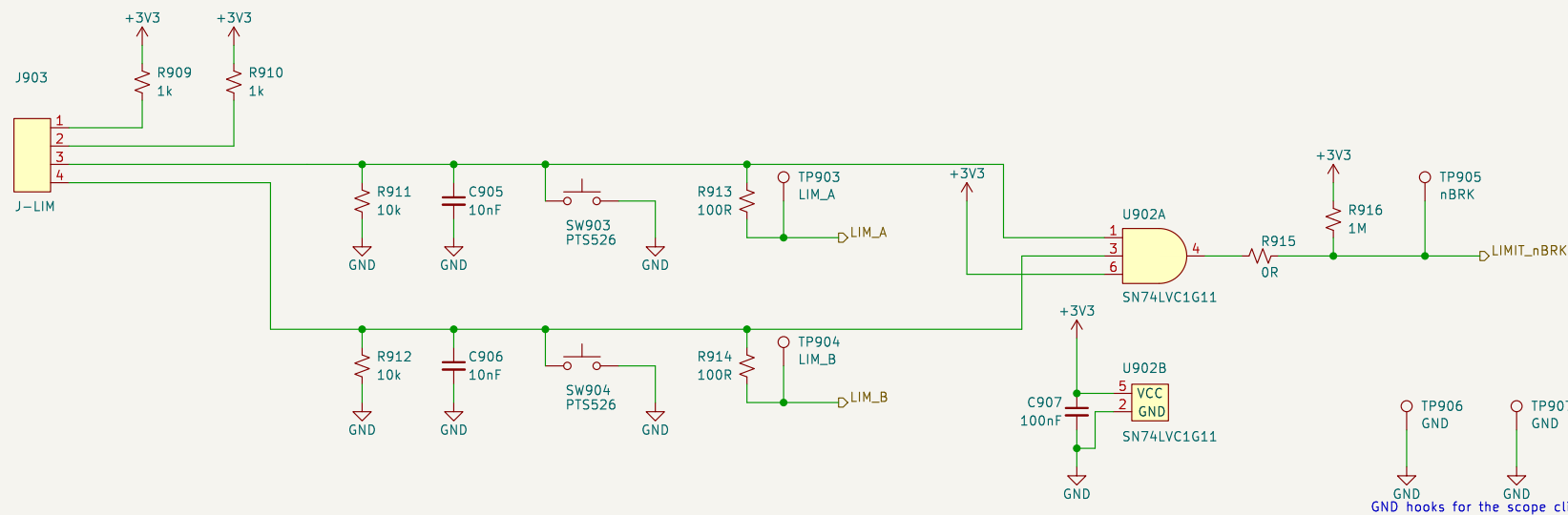
Active high, about 2.8 mA each.  
LED\_1 green is the heartbeat of  
TEST\_PLAN bring-up step 3; LED\_2  
orange is the fault / status lamp.

USER BUTTONS (BTN\_1 = HOME)



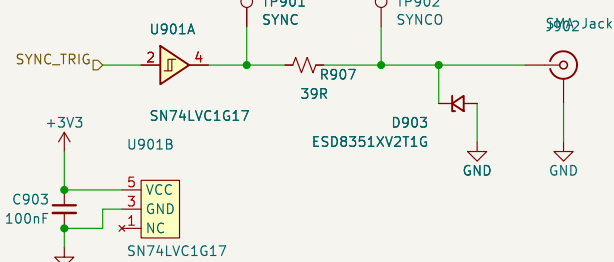
REQ-CC-06 asks for homing from the API OR an external button, so J901 parallels both on-board switches for a panel button. R905/R906 limit the contact current and set the debounce with C901/C902 – about 1 ms rise, 10 us fall. Which of BTN\_1 / BTN\_2 is HOME is OQ-05: firmware and silkscreen only, the two channels are electrically identical.

## END-OF-STROKE LIMIT SWITCHES + TIM1 BREAK



J-LIM pinout: 1 LIM\_A feed 2 LIM\_B feed 3 LIM\_A return 4 LIM\_B return.  
Both switches are NORMALLY CLOSED, so a closed switch pulls its node HIGH (3.0 V) and an open switch – or a broken wire, or an unplugged connector – lets the 10 k pull-down take it LOW. LOW therefore means ‘at that end of stroke, or the wiring has failed’, which is the safe direction to fail in.

SYNC / TRIGGER OUT (TIM15\_CH1)



REQ-EL-05 / REQ-EL-06: 3.3 V push-pull, 50 ohm source terminated, SMA jack. U901 buffers PE5 so the MCU pin never sees the cable and the source impedance is repeatable: R907 39 R plus the LVC output impedance (10–15 ohm typical) is about 50 ohm. TP901 is pre-termination, TP902 post; the SMA itself is the coaxial access of TEST\_PLAN 2.3. D903 is 0.55 pF and does not soften the edge.

BLOCK NOTES - ui\_io

SAFETY. SPEC 9.1 asks only for a firmware kill on limit actuation (REQ-SF-02). The block diagram and the .ioc both carry a second, independent hardware path (REQ-SF-05, DEC-00123): U002 ANDs the two safety limit inputs to the kill signal. U002 is a 74VHC00 inverter. The inputs are closed and both harness legs are intact. Any limit reached, any broken wire, an unplugged J-LIM or a dead 3V3 rail drives it LOW into TMI1\_BK1N2, and the PWM outputs go inactive with firmware taking no part. The J-LIM\_B each the MCU on their own EXTL pins for the firmware kill and telling the two ends apart. The AND gate's third input is tied high.

R915 is the TEST\_PLAN 3.3 link that lets the two TIM1 BREAK sources be exercised alone. It is fitted by default and must be marked on the silkscreen. With it lifted, R916 (1 M) holds LIMIT\_NBRK high so the DRV8323\_nFAULT path can be tested on its own; the firmware kill through LIM\_A / LIM\_B is unaffected either way. 1 M is weak enough that a tripped chain still pulls the net down to about 30 mV.

SW903 / SW904 are the 'assert each limit without the mechanics' means of TEST\_PLAN 4: pressing one pulls its node low through the 1 k feed (3.3 mA), which is exactly the state that end of stroke produces. They act independently because the two switch loops are independent – a series safety chain could not be stimulated one end at a time.

C905 / C906 (10 nF) with the -0.9 k source give about 9 µs of rise filtering and 100 us of fall; at the 20 mm/s of REQ-ME-03 that is 2 um of extra travel before the break asserts, against millimetres of switch overtravel. R913 / R914 protect the MCU pins. No discrete TVS is fitted on the limit lines: the harness stays inside the instrument and the RC plus the series resistor already bound the current into the MCU clamps. The SMA is the one externally exposed pin here, and it does get a clamp (D903).

+3V3 and GND are generated in power\_rails and enter this block through the hierarchy; nothing here drives them, and power\_rails owns their PWR\_FLAGS, so project ERC is clean at severity=all.

Test strategy: docs/TEST\_PLAN.md  
 REQ-EL-05/06/09, REQ-CC-06, REQ-SF-01/02/05  
 LIMIT\_NBRK PE6 -> TIM1\_BKIN2: hardware motor kill (DEC-0012)  
 Status LEDs, user buttons, SMA SYNC output, end-of-stroke limits  
**Amodo Design**  
 Sheet: /ui\_io/  
 File: ui\_io.kicad\_sch

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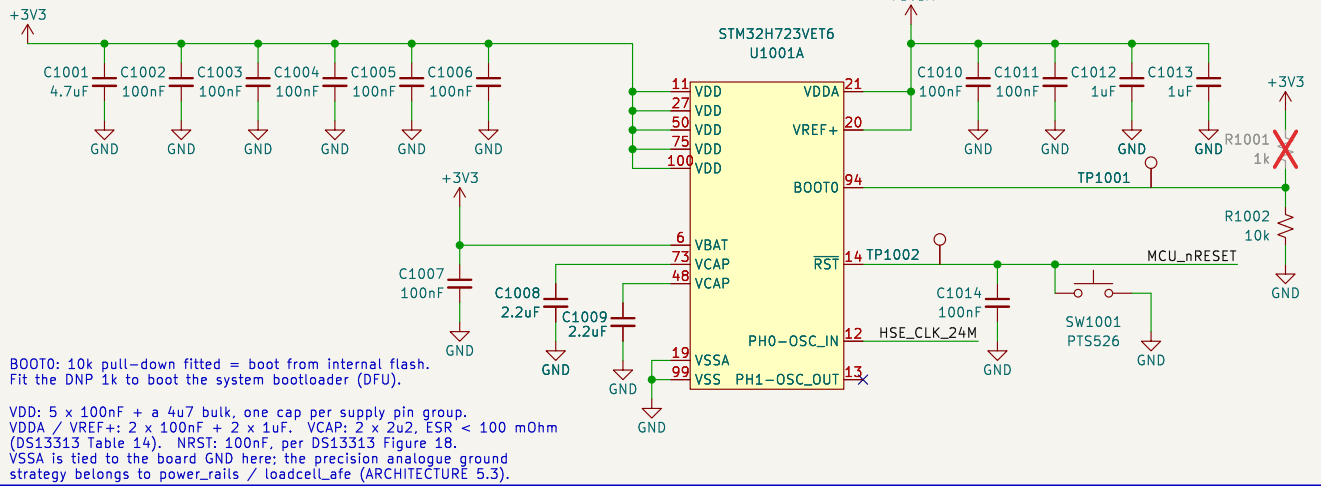
**Title: C Bs\_1 – User I/O, SYNC, Limit Switches**

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| Size: A3           | Date: 2026-09-02 |
| KiCad E.D.A. 9.0.8 |                  |

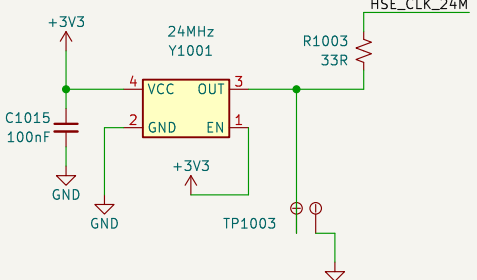


#### A MCU CORE – SUPPLIES, RESET, BOOT, HSE CLOCK IN



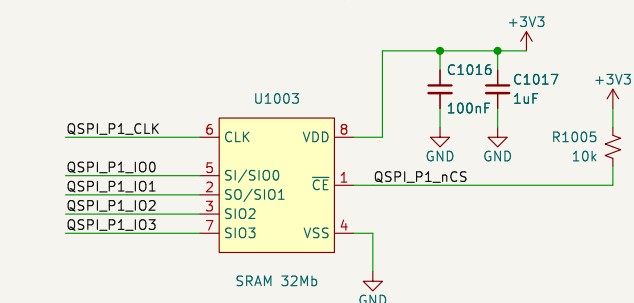
BOOT0: 10k pull-down fitted = boot from internal flash.  
Fit the DNP 1k to boot the system bootloader (DFU).  
VDD: 5 x 100nF + a 4u7 bulk, one cap per supply pin group.  
VDDA / VREF+: 2 x 100nF + 2 x 1uF. VCAP: 2 x 2u2, ESR < 100 mOhm (DS13313 Table 14). NRES: 100nF, per DS13313 Figure 18.  
VSSA is tied to the board GND here; the precision analogue ground strategy belongs to power\_rails / loadcellLafe (ARCHITECTURE 5.3).

#### B STM32 24 MHz HSE OSCILLATOR



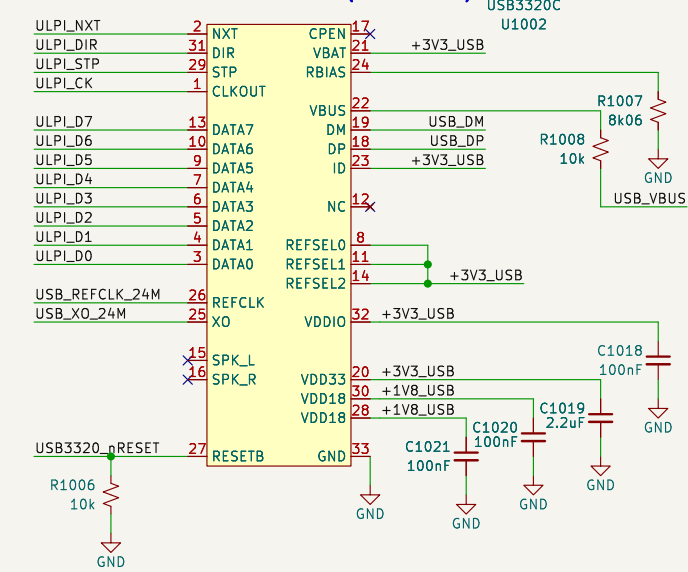
24 MHz LVCMOS MEMS oscillator, 10 ppm, driving the STM32 HSE ONLY – the USB3320 now has its own crystal (block C), so the two clock domains share nothing. STM32 HSE is an EXTERNAL 104 MHz, 50 ohm output driver; point to point, no series termination. CE# pull-up keeps the device deselected while the MCU pins are hi-Z. NO TEST POINTS ON THIS BUS – DEC-0018 ruling 1.

#### D OCTOSP1 FORCE-PROFILE RAM (QPI PSRAM) REQ-AR-12



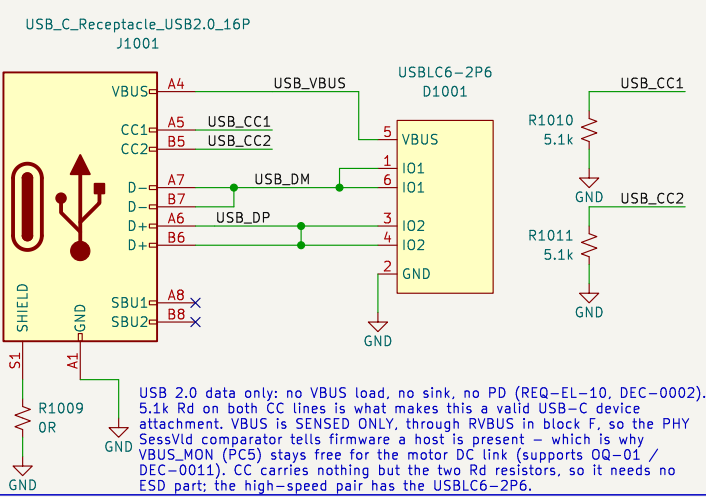
IS66WV54MBLL-104NLI : 32 Mbit (4M x 8) QPI PSRAM.  
ALL = 1.8 V part, BLL = 3.0 V part – BLL is the 3V3-compatible one.  
104 MHz, 50 ohm output driver; point to point, no series termination.  
CE# pull-up keeps the device deselected while the MCU pins are hi-Z.  
NO TEST POINTS ON THIS BUS – DEC-0018 ruling 1.

#### F USB3320C HI-SPEED ULPI PHY (REQ-AR-13)



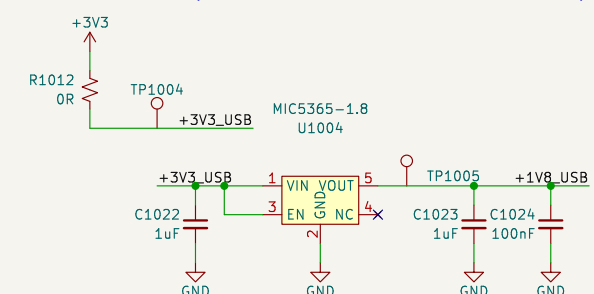
USB3320C-EZK-TR, ULPI OUTPUT CLOCK mode: the PHY generates the 60 MHz ULPI clock on CLKOUT -> STM32 ULPI\_CLK (PA5). Its 24 MHz reference is its OWN crystal on REFCLK / XO (block C) – no clock is shared with the STM32. REFSEL[2:0]=111 still selects 24 MHz (DS Table 5-10). RBIA5 8.06k 1% and RVBUS 10k (device-only) are datasheet-mandated values (DS 2.1, Table 5-7). ID tied to VDD33 = device-only operation (DS 5.6.1). CPEN / SPK\_L / SPK\_R / N-C are unused: no OTG VBUS switch, no audio. RESETB has no internal pull-down – the 10k holds the PHY in reset while the STM32 pins are hi-Z, which also covers the VDD18-before-VDDIO sequence note. NO TEST POINTS ON THE ULPI BUS – DEC-0018 ruling 1.

#### G USB-C RECEPTACLE – DATA ONLY, NO BUS POWER, NO PD



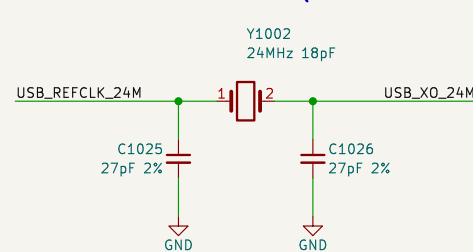
USB 2.0 data only: no VBUS load, no sink, no PD (REQ-EL-10, DEC-0002). 5.1k Rd on both CC lines is what makes this a valid USB-C device attachment. VBUS is SENSED ONLY, through RVBUS in block F, so the PHY sees VBUS comparator tells firmware a host is present – which is why VBUS\_MON (PC5) stays free for the motor DC link (supports OQ-01 / DEC-0011). CC carries nothing but the two Rd resistors, so it needs no ESD parts: the high-speed pair has the USBLC6-2P6.

#### E USB PHY RAILS (+3V3\_USB current break, +1V8\_USB LDO)



The USB3320 needs a real external 1.8 V supply on VDD18 (DS 2.1 pin 28/30 and DS 5.5.3) as well as 3.3 V on VBAT/VDD33/VDDIO, 1.8 V is generated here rather than requested from power\_rails, so no new cross-block rail is invented mid-wave: the mcu block therefore owns its test point and current break (producer-owns-the-break, DEC-0007 / TEST-PLAN 3). VBAT and VDD33 are tied together per USB3320 DS Figure 5-7 (3.3 V supply).

#### C USB3320 24 MHz CRYSTAL (ITS OWN REFERENCE)

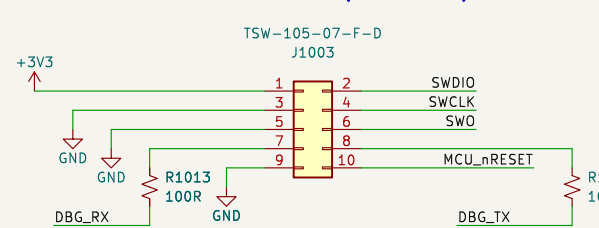


The USB3320 has its own 24 MHz reference: an ECS-240-12-23G-JGN-TR crystal on the PHY's internal oscillator between REFCLK and XO (DS Figure 5-4). It shares nothing with the STM32 HSE oscillator in block B. REFSEL[2:0] = 111 still selects a 24 MHz reference (DS Table 5-10), so no configuration changes.

Crystal: fundamental, parallel resonant, CL 18 pF, -40 to +85 C. DS Table 4-11 asks for CL 20 pF typ, ESR <= 30 ohm and a +/-500 ppm total budget – CONFIRM the ECS-23G ESR against 30 ohm before the first order.  
Load caps: C = 2 x (CL - Cstray) = 2 x (18 - 5) = 26 pF -> 27 pF C0G 2%, with Cstray = 3 pF pin capacitance (DS Table 4-11) + -2 pF PCB per side.

LAYOUT: crystal and both load caps within a few mm of pins 25 / 26, ground guard under the part, no other traces beneath it.

#### H SWD + USART3 DEBUG HEADER (REQ-AR-15)

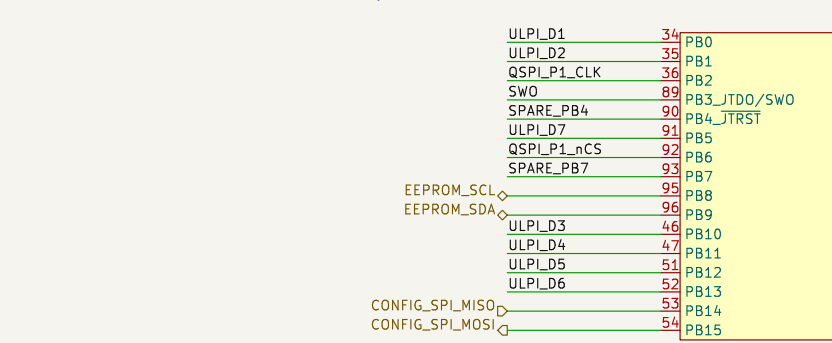
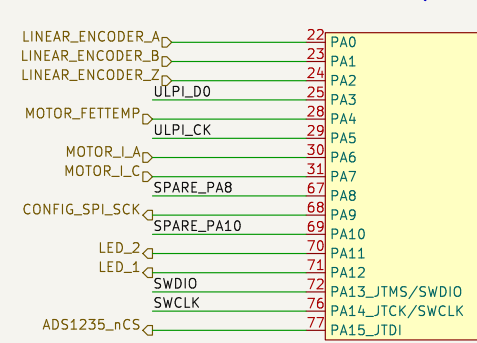


PINOUT – standard ARM Cortex Debug 10-pin order on 2.54 mm pitch, so a 2x5 SWD probe cable plugs straight on:  
1 Vtref (+3V3) 2 SWDIO 3 GND 4 SWCLK 5 GND  
6 SWO 7 DBG\_RX 8 DBG\_TX 9 GND 10 nRESET

Pins 7 and 8 are KEY / NC on the ARM pinout and are reused here for the USART3 console (REQ-AR-15 and the block diagram both put SWD and UART1 on ONE header). They are SWD-safe: an SWD probe drives only 2, 4 and 10 and reads 6, and the .ioc sets traces.Asynchronous\_SW, so PB4 / PB3 are not JTAG. The 100R series resistors make a mis-plugged JTAG probe or a shorted console pin a nuisance rather than damage.

DBG\_TX / DBG\_RX keep the .ioc net names: DBG\_TX is the STM32 OUTPUT (USART3\_TX, PD8), DBG\_RX the STM32 INPUT (USART3\_RX, PD9). MCU\_nRESET comes from block A, which owns its 100 nF, its push button and its test point. All six debug nets are sheet-local now that the header sits on the page it serves.

#### MCU PORTS – PIN ASSIGNMENT IS THE .ioc (DEC-0013); ULPI / QSPI / USB3320\_nRESET / SPARE\_\* stay on this sheet



## A



## C



## E



1



## 6

y i



## ch

sis

Requirements: docs/REQUIREMENTS.md    Tests: docs/TEST\_PLAN.md

Reasoning: docs/decisions/actuator-sch-motor.m...

3-phase inverter, phase-current sense, motor and rotary-encoder interface

**Amodo Design**

Sheet: /motor\_drive/

File: motor\_drive.kicad\_sch

**Title: CBs\_1 – Motor Drive (DRV8323S)**

Size: A3

Date: 2026-09-03

Rev: 0.2

KiCad E.D.A. 9.0.8

Id: 10/10